

CUSTOMER CHURNING PREDICTION USING
XGBoost & CATBoost
ABSTRACT

Businesses face intense competition for new customers, making client retention a critical focus. Early detection of client churn enables proactive strategies to maintain revenue and customer loyalty. This study investigates the efficiency of several machine-learning algorithms for early client churn prediction using a dataset encompassing customer data from the nine months preceding churn having 7043 rows (customers) and 21 columns (features). The proposed approach goes beyond previous work by implementing several techniques to enhance model performance and prediction accuracy.

These enhancements include oversampling of the training dataset using SMOTE to address class imbalance, outlier detection and handling to improve data quality, feature engineering through creation of new variables to capture more complex relationships, and dimensionality reduction using PCA to mitigate overfitting. Furthermore, gradient boosting algorithms such as XGBoost and CatBoost are explored, which are known for their high predictive accuracy and ability to capture intricate patterns within the data.

The study evaluates the performance of these enhanced algorithms alongside traditional methods like logistic regression, k-nearest neighbors, and random forest. The results show that the integration of data preprocessing and advanced gradient boosting models yields significant improvements in churn prediction accuracy, enabling businesses to target at-risk customers more effectively and optimize retention efforts. This research provides a robust and practical framework for businesses seeking to minimize customer attrition and maximize profitability through targeted and proactive customer relationship management.

Keywords: SMOTE, XGBoost, CatBoost, feature engineering, dimensionality reduction, PCA, data preprocessing, prediction accuracy, customer retention, gradient boosting.

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